

DOM Methods

Changing Styles with JavaScript

- Direct CSS Property Changes:
 - `document.getElementById("title").style.color = "blue";`
 - `document.querySelector(".btn").style.backgroundColor = "green";`
- Adding/Removing CSS Classes
 - `document.querySelector("#title").classList.add("highlight");`
 - `document.querySelector("#title").classList.remove("highlight");`

Separation of Concerns

- HTML: Content and elements (Structure)
- CSS: Visual appearance (Style)
- JavaScript: Interactivity and functionality (Behavior)

Text Manipulation Properties

- `innerText`: Text displayed on the page.
- `textContent`: Includes hidden content.
- `innerHTML`: Allows HTML content injection.

Text Content Example

- Example:
 - `let para = document.querySelector("#info");`
 - `para.innerText = "Updated Text!";`
 - `para.innerHTML = "<strong>Bold Text!</strong>";`

Manipulating Attributes

- Methods:
 - `getAttribute(attrName)`
 - `setAttribute(attrName, value)`
 - `removeAttribute(attrName)`

Attribute Example

- Example:
 - `let link = document.querySelector("a");`
 - `link.setAttribute("href", "https://example.com");`
 - `link.removeAttribute("target");`

What are Event Listeners?

- **Definition:** Event listeners are methods in JavaScript that allow you to detect and handle events triggered by user interactions or the browser.
- **Common Examples:**
 - Clicking a button
 - Hovering over an element
 - Typing in a text field
- **Key Concept:** Event listeners "listen" for a specific type of event on a specific DOM element.

Basic Syntax

- Code Example:

```
element.addEventListener(event, handler);
```

- **Parameters:**

- event: The type of event (e.g., "click", "mouseover", "keydown", etc.)
- handler: The function to execute when the event occurs.

Example 1 - Click Event

- **Scenario:** Change the background color of a button when it is clicked.
- Code Example:

```
const button = document.querySelector("#myButton");  
button.addEventListener("click", function () {  
    button.style.backgroundColor = "blue";  
});
```

Example 2 - Mouseover Event

- **cenario:** Highlight a menu item when the mouse hovers over it.
- Code Example:

```
const menuItem = document.querySelector("#menuItem");
menuItem.addEventListener("mouseover", function () {
    menuItem.style.backgroundColor = "lightgray";
});
menuItem.addEventListener("mouseout", function () {
    menuItem.style.backgroundColor = "";
});
```

Removing Event Listeners

- **Method:** removeEventListener
- Example:

```
function logMessage() {  
    console.log("Button clicked!");  
}  
button.addEventListener("click", logMessage);  
button.removeEventListener("click", logMessage);
```

- **Note:** The same function reference must be used for both addEventListener and removeEventListener.

Advanced Example - Keyboard Event

- **Scenario:** Detect when the user presses a specific key.
- Code Example:

```
document.addEventListener("keydown", (e) => {  
    if (e.key === "Enter") {  
        console.log("Enter key pressed!");  
    }  
});
```